(d) Find all possible values of m that make $z = \frac{\sqrt{3}+m\mathrm{i}}{1+\sqrt{3}\mathrm{i}}$ a purely real number.

$$\frac{\sqrt{3}+mi}{1+\sqrt{3}\,i} \times \frac{1-\sqrt{3}\,i}{1-\sqrt{3}\,i}$$
$$= \frac{\sqrt{3}-3i+mi-\sqrt{3}\,m}{4}$$
$$(m-3)i = 0$$
$$m = 3$$

(e) If $|z| = 1$, and $z \neq 1$, prove that $\frac{1+z}{1-z}$ is purely imaginary.

$$z = x + yi$$
$$\sqrt{x^2+y^2} = 1$$
$$x^2 + y^2 = 1$$